

SOL - practice sql - hospital database

- 1) Use '=' operator after WHERE when you are actually comparing with real values.
- 2) Use 'IS' when we just want to check missing values
Eg → ~~not~~ NULL, IS NULL
- 3) ~~less~~ LIKE operator - used when we want to match a pattern rather than an exact value.

a) % - means any no. of characters (including zero)
Eg → WHERE first-name LIKE 'C%';
This says that first character must be C, and then anything can come after.

ends with n - Eg → WHERE first-name LIKE '%n';

contains 'ae' → Eg → WHERE first-name LIKE '%ae%';

b) How to find exactly 5 characters? - underscores

→ WHERE first-name LIKE '-----';
(has exactly 5 characters)

→ WHERE first-name LIKE 'C-----';
(has 5 characters and starts with C)

Note ⇒ LIKE is used to match patterns while '=' is used to match exact values.

→ WHERE first-name LIKE '_a%';
(name whose second letter is a)

→ 'a%.n' (name beginning with a & ends with n)

4) IN operator - can group multiple equals to op

Eg → WHERE city = 'DELHI'
OR " = 'Mumbai'
" " = 'Pune';

} WHERE city in
('Delhi', 'Mumbai',
'Pune');

d) BETWEEN
Instead of WHERE age ≥ 20 AND age ≤ 20 ; (This is exclusive)
USE $\rightarrow$ WHERE age BETWEEN 20 AND 20; (This is inclusive) accordingly

NOTE $\rightarrow$ Don't use LIKE directly after table name, we need to use WHERE before that.

e) UPDATE $\rightarrow$ when we want to modify some data.

Syntax $\rightarrow$ UPDATE table-name SET column-name = new value WHERE <condition>;

$\rightarrow$ CASE $\rightarrow$ when diff. rows need different updates.

Syntax $\rightarrow$ UPDATE table-name SET column-name CASE
WHEN cond1 THEN val1
" cond2 " "2
ELSE column-name
END;

Note - don't use where clause when you want to update all rows.

4 Main SQL commands

- 1) Select
- 2) Update
- 3) Insert

Syntax $\rightarrow$ INSERT INTO table-name (col1, col2) VALUES (val1, val2);

We can also add multiple rows at once, by providing multiple values.

8) CONCAT

Syntax $\rightarrow$ SELECT CONCAT (first-name, last-name)
FROM table;

Can also give this column a new name as keyword.

It will create a new column in table, will show an a/p.

9) AS → aliases

10) UPPER() → Converts to uppercase

Eg → SELECT UPPER(first-name) FROM patients;

11) LOWER() → Converts to lower case.

12) LENGTH() → returns the no. of characters.

Eg → SELECT LENGTH(first-name) FROM patients;

13) TRIM() → removes spaces from start and end.

Eg → SELECT TRIM('John');

Note → We can nest diff operators. Eg → LOWER(CONCAT

14) To select from two tables without combining them,

Eg → SELECT * FROM patients, doctors;

This creates a cartesian prod.

15) JOIN

Humo jo info first krai hai wo do diff tables mai hai therefore we need to join tables.

Stor pehle hum dono tables mai ek common attribute dhundenge, which it will be the joining key. (Here its province-id)

Syntax → SELECT columns FROM table1, JOIN table2
ON table1.column-common = table2.column-common

More professional way to write it →

SELECT p.first-name, p.last-name, pn.province-name

FROM patients p JOIN province-names pn ON

p.province-id = pn.province-id.

→ We need join when some info is in some other table.

16) COUNT() → Select COUNT(*)

17) YEAR(date) → extracts the year part of the date.

18) MONTH(date) or DAY(date)

19) MAX(column name) → selects max value in that column